Climate preconditioning and the mechanics of hazardous alpine mass movements: a catalog and attribution protocol

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Abstract

Large rock, ice, and mixed-material collapses can transform into far-traveled debris flows or locally extreme tsunamis. Warming changes glacier geometry, permafrost temperature, meltwater supply, and precipitation phase. However, event reports seldom separate these preconditioning changes from immediate weather, earthquakes, structure, and progressive damage. Here, I define a climate-blind event inventory and an attribution protocol that preserves this distinction. Two structured frames contain 40 and 38 rows that cross an operational volume, travel, or runup prefilter. An exact provenance crosswalk gives every row a visible disposition, and supplemental searches extend the screen to 148 candidate occurrences. Two independent screens include 73, retain 68 as uncertain, and exclude 7; 53 included failures have day-scale or better timing for a later trigger analysis. Trigger, cascade, and repeat-site dependence remain separate. A registered ERA5 pilot then retains the 29 of 53 included, trigger-time-eligible occurrences that have finite catalog coordinates. For the registered seven-day pre-date window, the median rank among 1991–2020 calendar controls is 0.586; after a robust linear trend is removed, the median residual rank is 0.483. A conservative-antecedence sensitivity gives 0.586 and 0.533. The median four-cell rank ranges are 0.133 and 0.103, respectively. An independently reviewed source-coordinate and UTC-onset audit subsequently accepts source coordinates for 24 occurrences, UTC intervals for 34, and both for 22; only 9 of the pilot's 29 occurrences have accepted source coordinates. The registered pilot results are preserved rather than recomputed post hoc. These remain descriptive exposure diagnostics because no denominator is yet aligned with the cases. A separate climate- and case-blind terrain pilot enumerates glacier, glacier-contact, and permafrost-intersection objects in three deterministically selected windows. Primary connected-component counts change by factors of 1.3–3.0 between 30 and 90 m grids, so the pilot supplies a reproducible frame but not a unique population count. I then specify detrended seasonal controls, matched non-failing slopes, mechanism-specific glacier and permafrost observations, and multiplicity control. Uneven observation and unresolved primary sources prevent these counts from estimating a frequency trend. No systematic climatic association or causal effect is estimated here.

1 Introduction

Alpine mass movements become especially hazardous when an initial failure changes material or enters a confined channel. The 2021 Chamoli rock and ice avalanche transformed into a debris flow and flood that scoured valley walls and struck two hydropower projects [19]. Fjord failures instead transfer momentum to water. The September 2023 Dickson Fjord rockslide generated near-source runup of about 200 m and a very-long-period signal that persisted for nine days [16, 21]. The August 2025 Tracy Arm landslide produced a measured maximum runup of 481 ± 7.6 m [20]. A useful catalog must describe the initial failure separately from the downstream cascade.

The mechanical origin of the initial failure remains the central uncertainty. Warming may remove glacier support from a rock slope, thaw ice in fractures, or change meltwater pressure beneath a glacier. Weather may provide a shorter perturbation through rain, snow loading, melt, or rapid temperature change. Lithology, adverse structure, progressive damage, valley geometry, glacier dynamics, and earthquakes remain competing or interacting explanations. A climatic trend can therefore precondition a failure without being its immediate trigger, and a warm event year does not establish causation.

Here, I use a literature seed to define a reproducible catalog and the tests that would be needed to evaluate systematic association with climate. The seed is a protocol-development set, not an inference sample. A separate, climate-blind discovery inventory supplies the candidate sampling frame. The two are kept separate so that examples chosen because they motivated a climate hypothesis cannot enter an association test without being rediscovered under the same rules as every other event.

2 Catalog construction and sampling protocol

The catalog consists of three linked tables. The event table records identity, group, analysis role, date, location, setting, initial failure, and downstream transformations. Dates and coordinates have field-specific sources. The measurement table stores one quantity per row and distinguishes a point value, approximation, lower bound, upper limit, or range. It records uncertainty type, observation period, scenario, evidence kind, and source locator separately. The claim table labels each statement as observation, preliminary observation, reconstruction, model output, published inference, or project interpretation. Each event has explicit process-chain, immediate-trigger, and climate-preconditioning claims, including an unresolved claim when evidence is absent.

The registered geophysical inclusion rule requires rapid failure of rock, glacier ice, or a mixed mass in an alpine or fjord setting, a documented time and source location, and at least one of three operational thresholds: initial volume of 10^6 m^3, source-to-deposit travel of 2 km, or landslide-tsunami field runup of 10 m. Downstream damage and population exposure are stored as consequences but do not determine geophysical inclusion. Snow avalanches without glacier-ice or rock failure and floods without an initial mass movement are outside the scope. These thresholds were frozen before the climate-blind screen. A separately labeled active slope may enter the protocol-development set when measured deformation and published failure geometries exist. Barry Arm is such a prospective case, not an occurred event [1–3, 18].

Discovery covers 1 January 2000 through 30 August 2026. Two enumerated frames are the 354-case PANGAEA landslide-tsunami catalog [4] and the 60-event High Mountain Asia rock- and ice-avalanche inventory [25, 26]. A 51-event global review supplies a cross-check [15]; its linked spreadsheet was not enumerated because automated retrieval returned HTTP 403. Independent query suites for fjord and lake tsunamis, glacier failures, and rock or mixed avalanches supplement dates and regions not covered by the structured frames. All exact queries, execution times, and resource classes are archived with the catalog. Searches and candidate fields omit asserted climate cause and event climate exposure. Inventory cause columns are ignored.

Every candidate promoted to detailed screening has two differently named reviewers. Each reviewer sees identity, time, place, initial material and process, the threshold observation, and its source. Allowed decisions are include, exclude, or uncertain. An inventory value can discover an event, but final inclusion requires a primary event reconstruction or authoritative agency record. A secondary-only record remains uncertain. Conflicting dates, coordinates, volumes, and aliases remain in the notes rather than being silently reconciled. Detection era, sensor coverage, and reporting source must remain in the later observation model because discovery is not stationary.

Table 1: Cause-blind flow through the two enumerated discovery frames. "Retained" applies the occurrence window and a usable event record; the numeric prefilter uses only published volume, travel, or runup fields. It does not establish material or setting eligibility.

Frame	Published	Downloaded	Window	Retained	Numeric
PANGAEA landslide tsunamis	354	355	87	84	40
High Mountain Asia	60	60	40	40	38

The reconciliation crosswalk links all 78 numeric-prefilter rows in Table 1 to candidate records. The source bytes, checksums, and qualifying-row manifest are archived with the catalog. Compound inventory rows are expanded into individual occurrences. A row-level size estimate is not assigned to each component unless the primary source supports that allocation; otherwise each component remains uncertain. Independent searches identified, for example, the November 2025 Bartuytsete ice-rock avalanche, whose published 3.5 km runout passes the travel threshold [5]. The resulting table contains 148 candidates: 73 included, 68 uncertain, and 7 excluded. Fifty-three included failures have day-scale or better timing and enter the future trigger-time risk set; the remainder can inform only site-level or preconditioning analyses.

The seed records were selected because they motivated the study and still cannot support a frequency trend, an event-conditioned climate distribution, or a global susceptibility model. The discovery table is closer to an inference frame, but it also cannot support those estimates until the registered search cutoff, both screens, disagreement resolution, and catalog commit are frozen. No atmospheric anomaly or cryospheric exposure is extracted before that freeze.

The protocol has two estimands. The episode-level trigger estimand compares conditions at failure with time controls; the site-level preconditioning estimand compares source slopes with an incidence-density risk set. The two Aru glaciers are separate episode candidates but share a site-level cluster. Distinct Aysen, Denali, and Kunlun source failures remain individual events even though each set shares an earthquake trigger. The unresolved October 2023 Dickson Fjord episode is linked to the September site but receives no analysis role until its source-specific threshold is established. Barry Arm is excluded from event-time inference because catastrophic failure has not occurred; it instead supplies a prospective case for testing susceptibility variables.

3 Mechanics represented by the seed records

The two Aru glacier tongues detached after surge-like acceleration on low-angle terrain; the published mean Aru-1 surface slope is approximately 13°. Remote sensing and thermomechanical modeling indicate that reduced basal friction shifted resistance toward frozen tongues and lateral margins until those forces were exceeded [10, 14]. Gilbert and colleagues inferred a five- to six-year increase in melt and rain rather than exceptional seasonal forcing. For Aru-1, approximately 10–25 mm of precipitation during the two days before failure is a possible short perturbation; Aru-2 has no resolved immediate trigger. The two collapses traveled approximately 6 and 5 km beyond their respective glacier termini [14].

At Chamoli, 26.9×10^6 m^3 of rock and glacier ice failed from Ronti Peak and transformed into a highly mobile debris flow [19]. The reported 95% reconstruction interval is 26.5–27.3×10^6 m^3. The mass was about 80% rock and 20% ice; these are model-dependent composition estimates rather than direct material measurements. Mapping and reconstruction do not require a glacial-lake outburst to explain the flood cascade. Geologic structure and progressive damage remain alternatives to thermal and hydrologic preconditioning. Snow and ice loading on the headwall and a preceding positive temperature anomaly have been proposed as final perturbations, but a unique immediate trigger was not isolated [24].

At Dickson Fjord, a foliation-parallel rockslide struck a gully glacier before entering the fjord. The reconstructed rock volume was $25.5 \pm 1.7 \times 10^6$ m^3, with a separate $2.3 \pm 0.15 \times 10^6$ m^3 entrained-ice estimate [21]. Rockslide, glacier impact, tsunami, and the very-long-period signal are observations or reconstructions. The cross-fjord seiche was first inferred by wave and seismic modeling; sparse SWOT altimetry later provided direct water-level observations consistent with its phase and period [16]. Multidecadal glacier lowering supports a debuttressing hypothesis, but no proximate trigger or formal anthropogenic event attribution was demonstrated.

Tracy Arm and Barry Arm illustrate occurred and prospective consequences of glacier retreat along fjords. The mapped Tracy Arm volume gives a lower limit of 63.5×10^6 m^3 with a reported uncertainty of 2.7×10^6 m^3; unmeasured submarine failure may increase the total. Several days of accelerating microseismicity preceded failure [20]. Regional summer warming since 1875 was quantitatively attributed to anthropogenic forcing. Observed retreat of South Sawyer Glacier supports debuttressing as a mechanical inference, but the glacier retreat itself was not assigned an anthropogenic counterfactual fraction. At Barry Arm, slope displacement and glacier retreat were temporally coincident during 2010–2017 [3]. Their reported correlation of 0.99 compares cumulative trending series and is not by itself a causal test. Published geometry scenarios span $290–689 \times 10^6$ m^3 [1], while a four-layer seismic inversion estimates a local mean failure-surface depth of 188 ± 9 m [2]. These are uncertain slope geometries, not a failed volume.

Table 2: Seven focal records in the protocol seed. The dependent October 2023 Dickson Fjord episode remains in the machine-readable catalog but is omitted here. Volumes describe an initial failed mass except at Barry Arm, where the range describes possible unstable-slope geometries. Shown uncertainties are reproduced from the sources; their distinct confidence, conservative-sum, or unspecified semantics are stored in the machine-readable catalog.

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Nepal, 26 Aug 2026	Provisional glacial-collapse classification	unresolved	Debris flow and flood traveled about 100 km	No event-specific attribution; process reconstruction remains provisional [22, 23]
Chamoli, 7 Feb 2021	Rock and glacier ice	26.9 [26.5, 27.3]	Avalanche to debris flow and flood	Several thermal effects are plausible; no single trigger isolated [19, 24]
Aru-1, 17 Jul 2016	Glacier detachment	68 ± 2	Ice avalanche; 6 km beyond terminus	Multiyear melt and rain inferred to increase subglacial water pressure [10]

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Aru-2, 21 Sep local time 2016	Glacier detachment	83 ± 2	Two pulses; 5 km beyond terminus	Shared multiyear preconditioning; immediate trigger unresolved [10, 14]
Dickson Fjord, 16 Sep 2023	Foliation-parallel rockslide	25.5 ± 1.7 rock	Glacier entrainment, tsunami, and nine-day signal	Glacier thinning inferred to debuttress the toe; proximate trigger unresolved [21]
Tracy Arm, 10 Aug 2025	Bedrock landslide	≥ 63.5 mapped	Glacier/fjord impact, tsunami, and signal lasting at most 36 hours	Regional warming attributed; retreat–debuttressing link is a mechanical inference [20]
Barry Arm, active	Slow bedrock landslide	290–689 possible	Potential rapid failure and tsunami	Retreat and motion are temporally coincident; alternatives require testing [1, 3]

4 Mechanical predictors beyond reanalysis

Atmospheric fields do not measure glacier support, permafrost fracture state, or basal effective pressure. The analysis will therefore pair reanalysis with annual optical and radar mapping of terminus position, glacier-slope contact, surface elevation, and source-slope deformation. Define Δh_i as index-date thickness minus baseline thickness at the slope contact. A signed stress-scale proxy for support loss is

$$S_i = -\rho_i g \Delta h_i, \tag{1}$$

where ρ_i is ice density, so thinning gives positive S_i and thickening gives negative S_i. This proxy has stress units but is not the lateral buttressing traction. Geometry-specific elastic or limit-equilibrium calculations are required to translate it into a support change. Results will therefore report signed thickness change, its magnitude, mapped contact loss, and S_i before interpreting debuttressing.

For a first glacier-detachment screening bound, treat the bed as Coulomb till:

$$\begin{aligned} N &= p_i - p_w, \quad & \tau_y &= c + N \tan\phi, \\ \tau_b &\leq \tau_y, & \tau_d &\simeq \rho_i g h \sin\alpha, \end{aligned} \tag{2}$$

where N is effective pressure, p_i is ice overburden pressure, p_w is water pressure, c and ϕ are effective till cohesion and friction angle, h is ice thickness, and α is surface slope. In a slab that neglects lateral and longitudinal resistance, basal yield cannot balance driving stress when

$$N < N_{\text{crit}} = \frac{\tau_d - c}{\tan\phi}. \tag{3}$$

This is a conditional screening range, not an inferred Aru effective pressure; the analysis must span plausible c, ϕ, geometry, and neglected margin stresses. Melt or rain can raise p_w and reduce resistance when drainage is inefficient. The same input can enlarge efficient drainage, lower p_w, and increase resistance.

Permafrost hypotheses require rock-temperature or fracture-ice observations, not air temperature alone. Where boreholes are absent, the protocol will use modeled rock-surface temperature, topographic shading, fracture orientation, and cumulative thermal history as proxies and will label the inference. Adverse structure is a baseline confounder; deformation may be a mediator or competing-process observation. The causal specification will determine which variables are adjusted rather than treating every measured field as a covariate.

4.1 Climate- and case-blind susceptible-terrain pilot

A denominator for an incidence calculation must be constructed without using the failures it will later explain. I therefore registered a terrain pilot before reading source-product values. It fixes RGI 7.0 regions 01, 11, and 13 and chooses one 1° window per region using only catalog fields [17]. The selector uses the archived `cenlon` and `cenlat` points, which the RGI metadata describes as approximately central points within the outlines rather than geometric centroids. Eligible cells contain at least 10 such points and 1 km^2 of summed glacier area; the cell with the smallest SHA-256 digest of its region and integer edges is selected. This rule gives 58–59°N, 155–154°W in Alaska; 45–46°N, 7–8°E in Central Europe; and 43–44°N, 94–95°E in Central Asia. These are validation windows, not regional samples.

RGI polygons define glacier objects and the ice mask. Native Copernicus DEM GLO-30 values are warped to a local Lambert azimuthal equal-area grid in each window [8]. Centered differences define surface slope at 30 m; exact 3×3 elevation means give the aligned 90 m sensitivity. Outside-glacier terrain is joined by four-neighbor connectivity. The glacier-contact stratum requires slope of at least 30° and raster edge distance no greater than 100 m. The permafrost stratum applies the same slope threshold and intersects PZI ≥ 0.1 from the 30-arc-second global zonation map [11, 12]. PZI is a climatological index based on 1961–1990 air temperature, not a measured local temperature or failure probability. The Global PZI data source is the World Glacier Monitoring Service (WGMS), with Gruber as creator and UZH as publisher. The machine-readable tables retain slope thresholds of 25, 30, and 35°, contact distances of 0, 100, and 300 m, PZI thresholds of 0.1 and 0.5, and both grid spacings. ITS_LIVE version 2 contributes only its 2014–2022 image-pair count as an observability field; no velocity or trend enters object membership [9].

Counts below include boundary-truncated objects; the archived summary also reports wholly contained objects separately.

For every object of area A, the calculation reports $V_d = Ad$ for prescribed depths $d = 10$, 30, and 100 m. An object is conditionally eligible when $V_d \geq 10^6$ m^3. This does not estimate glacier thickness, rock-failure depth, or mobilized volume. The primary 30 m calculation contains 140, 472, and 144 glacier polygons in Alaska, Central Europe, and Central Asia, respectively. The corresponding glacier-contact counts are 1,766, 2,872, and 742; permafrost-intersection counts are 3,672, 5,321, and 7,099 (Figure 1). At $d = 30$ m, conditional counts are 137, 340, and 117 for glaciers; 331, 611, and 220 for glacier-contact terrain; and 724, 753, and 1,086 for permafrost-intersection terrain. The three strata overlap and these values must not be summed.

Area is less sensitive than object count. From 30 to 90 m, glacier-contact area changes by −8%, +6%, and +3% in the three regions, whereas its component count decreases by factors of 1.7, 1.6, and 1.3. Permafrost-intersection area decreases by 18%, 11%, and 35%, while component count decreases by factors of 1.9, 2.0, and 3.0. Conditional counts need not vary monotonically: coarsening removes steep pixels but also joins patches. Increasing contact distance similarly increases total area while it can reduce the number of volume-eligible components by merging them. A deterministic audit of five objects per stratum and window reproduced every source-mask predicate, area, connectivity, boundary flag, and coverage field.

4.2 Held-out susceptible-area convergence

The object-count sensitivity above motivates a separate test of area as a possible exposure measure. Before reading new inventory values, I ranked 14 RGI region codes by SHA-256 and fixed regions 07, 08, and 03. A second blind hash selected 77–78°N, 18–19°E in Svalbard; 69–70°N, 21–22°E in Scandinavia; and 81–82°N, 85–84°W in Arctic Canada North. The 200 eligible window keys and their digests were frozen before a DEM or PZI value was read. These deterministic windows are held out from the pilot, but they are not a random sample of mountain terrain.

The registered reference is total area under the pilot's 30° slope, 100 m glacier-contact, and PZI ≥ 0.1 screens. I independently warp the DEM and masks to four 30 m grids separated by half-cell phase shifts. The 90 m variant averages complete 3×3 elevation blocks but samples masks on its own grid. A stratum passes only if every window has a 90 m area departure no greater than 20% and a population coefficient of variation no greater than 0.10 across the four 30 m phases.

Grid phase has little effect (Figure 2). The largest phase coefficient of variation is 0.00375. In contrast, coarsening reduces glacier-contact area by 26.7% in Arctic Canada and 36.6% in Svalbard; the change is 0.7% in Scandinavia. PZI-intersection area decreases by 29.1%, 37.4%, and 14.2%, respectively. Both strata therefore fail the registered all-window criterion. This comparison suggests that half-cell alignment is not the controlling numerical uncertainty among the tested variants. Resolution coarsening changes DEM smoothing, slope support, mask sampling, contact-distance discretization, reporting centers, and cell area together, so this test does not isolate which operation causes the departure. Neither area measure is used below to infer a global incidence density or hazard distribution.

The registered geometric choices also remain influential. Across the three windows, the unshifted-reference-grid 30 m sensitivity calculations span 0.27–114.0 km^2 for glacier-contact area and 13.6–403.4 km^2 for PZI-intersection area as slope, contact distance, and PZI threshold vary. These ranges describe the screens; they are not uncertainty intervals or alternative opportunities to pass the convergence test.

5 Event-centered climate test

For each frozen event candidate, reanalysis will provide near-surface air temperature, precipitation phase and amount, snow accumulation, radiative and turbulent fluxes, and surface pressure. Prespecified reductions will include cumulative positive degree days, liquid-water input, recent snow load, freeze–thaw crossings, and a multidecadal temperature trend. Windows, directions, and mechanism-specific primary contrasts will be registered before event values are examined.

Immediate-trigger anomalies and long-term preconditioning require different controls. For trigger variables, I will remove the local seasonal cycle and a smooth long-term trend using data outside the event window. The event residual will be compared with time-matched, same-season blocks. Block resampling will preserve serial correlation, and the two Aru records will share a site-level cluster. This avoids a null that ranks recent event years against an earlier, cooler climate and then interprets the resulting rank as unusual weather.

For persistent preconditioning, the comparison will use incidence-density risk sets. At each event date, eligible controls are catalog-search source slopes that remain under observation and have not yet failed. Controls receive the case index date, and every exposure window ends before that date; a

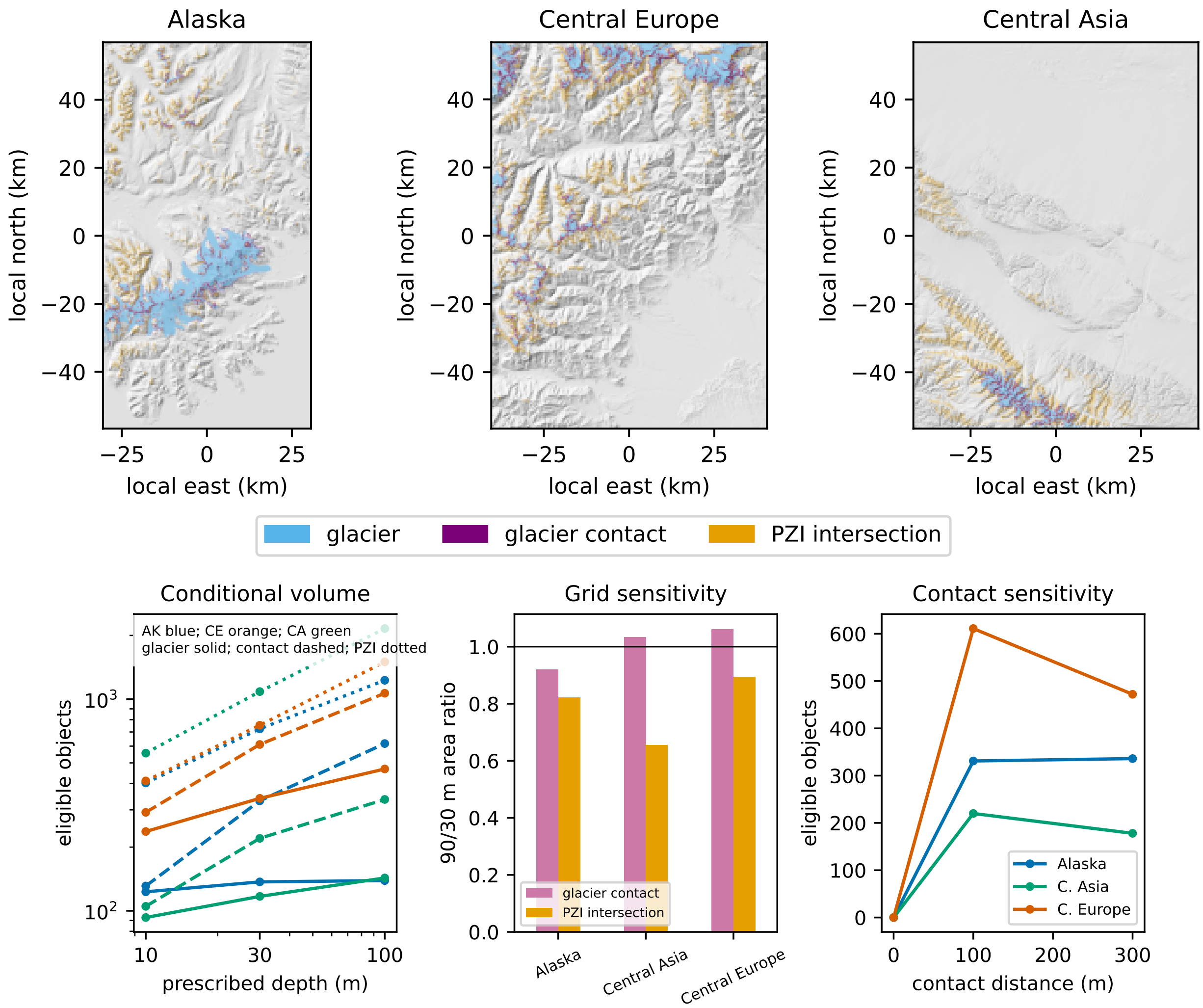

Figure 1: Climate- and case-blind susceptible-terrain pilot. Top: the three deterministically selected windows with RGI glacier polygons, primary steep glacier-contact terrain, and primary steep PZI-intersection terrain. Bottom left: conditional-volume counts across prescribed depths. Bottom centre: primary rock-slope 90/30 m area ratios. Bottom right: conditional counts across glacier-contact distances. Counts include boundary-truncated objects. In the depth panel, colors distinguish regions and line styles distinguish strata. Strata overlap, source epochs differ, and none of the mapped objects is a predicted failure. Elevation products were produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved.

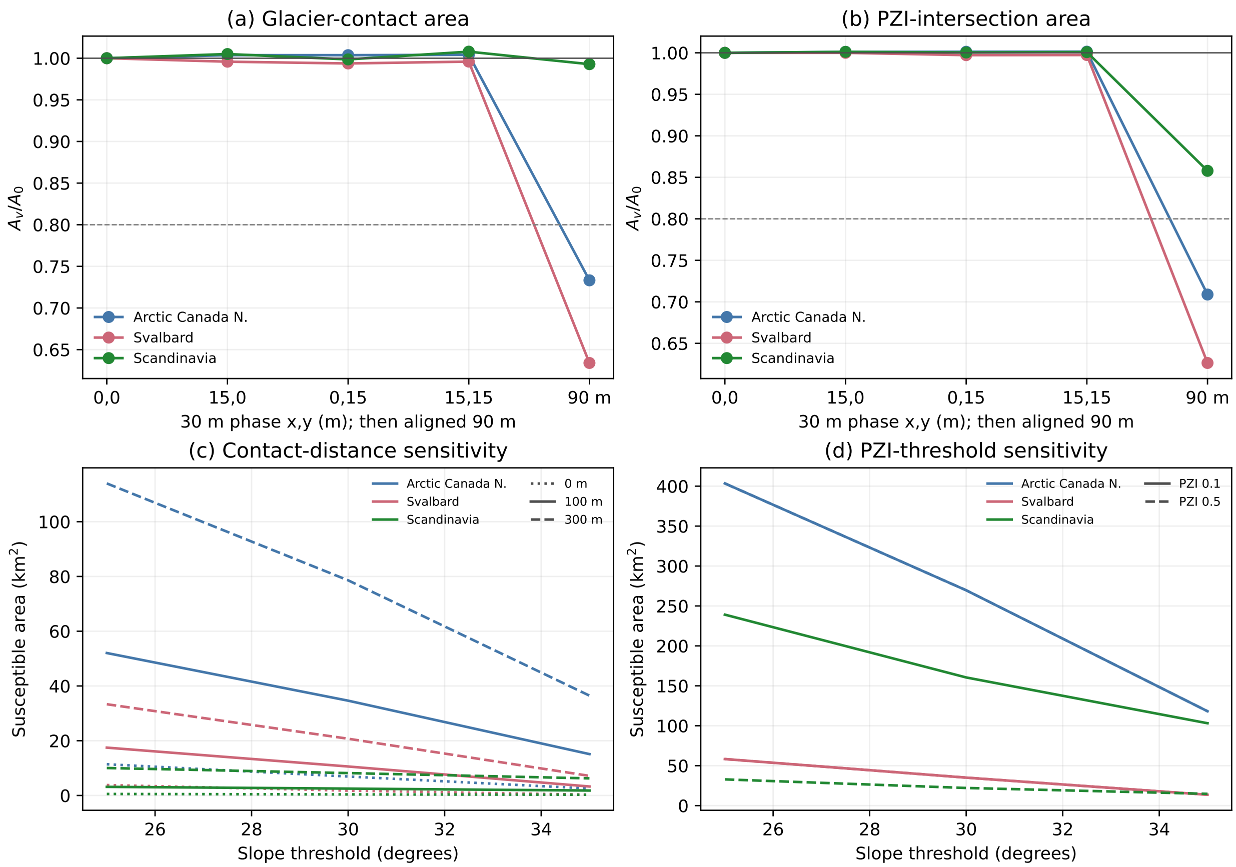

Figure 2: Held-out susceptible-area convergence. Panels (a) and (b) show area relative to the unshifted 30 m result for three half-cell phase grids and the aligned 90 m grid. The dashed 0.8 line is the lower registered resolution bound and applies to the 90 m point; phase is assessed by its four-grid coefficient of variation. Panels (c) and (d) show separate 30 m sensitivity to slope, contact distance, and PZI threshold. Colors distinguish held-out regions and line styles distinguish geometric thresholds. The plotted objects are not predicted failures. Global PZI data are attributed to WGMS as source, Gruber as creator, and UZH as publisher. Elevation products were produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved.

control may later become a case. Follow-up ends at failure, loss of adequate sensor coverage, or the registered study end. Matching uses baseline region, elevation, aspect, relief, lithology where available, and pre-exposure sensor coverage. It does not use index-date glacier contact or deformation, which may lie on the causal pathway. A preregistered causal diagram will distinguish baseline confounders from exposures and mediators.

A hierarchical model will estimate associations while including region and site-group effects, detection era, and monitoring intensity. Glacier-detachment sites will have a signed liquid-water/effective-pressure contrast; glacier-contact rock slopes will have a signed contact-loss/support contrast; and permafrost rock slopes will have a cumulative thermal contrast. Predictors will be standardized against their own risk-set distributions and will not be imputed across inapplicable mechanism classes. Family-wise error will be controlled across these primary contrasts; secondary predictors will use a false-discovery-rate procedure. An association under this design would still not constitute a complete anthropogenic event attribution.

Global products smooth the elevation and aspect that govern steep-terrain energy balance. Downscaling will begin with corrections tied to a named bias: temperature adjusted between grid-cell and source elevation using a documented lapse rate, precipitation separated by phase, and shortwave radiation corrected for slope, aspect, and shading. Results will be reported before and after each correction. Regional modeling is warranted only when these corrections change an event anomaly enough to affect the prespecified test.

5.1 Source-coordinate and UTC-onset audit

After the registered temperature pilot exposed location and clock ambiguity, I registered a separate, climate-blind audit before retrieving any new event exposure. It fixes the same 53 trigger-time-eligible occurrences and preserves every original catalog value. The coordinate target is the initiating source or source-area centroid, not a named summit, deposit, lake, gauge, dam, or downstream impact. Each assertion records geometry role, evidence method (published, digitized, gazetteer, or unavailable), a horizontal-uncertainty class that may be unbounded, exact source locator, and reviewer state. Digitized points are accepted only when map control supports the registered uncertainty class.

Time assertions target the earliest documented motion of the qualifying source failure and are stored as half-open UTC intervals. Local clocks without a documented civil scale are not converted from longitude. Multi-pulse events retain the qualifying pulse range when source volume cannot be allocated by pulse. An earthquake origin can be retained only after an event-specific source establishes that the named failure was coseismic; it remains labeled a trigger-origin proxy rather than a measured detachment clock. Two differently named reviewers check each assertion. The final tables, validator, protocol, and retrieved legally redistributable source objects are checksummed in a manifest frozen before any new reanalysis extraction. Seven otherwise reusable objects could not be archived because automated retrieval was blocked; their access failures remain explicit in the source table.

The audit accepts source coordinates for 24 of 53 occurrences: 21 have uncertainty bounded at 1 km and 3 at 5 km. It accepts UTC intervals for 34, comprising 13 minute, 11 second, and 10 range assertions; 3 more retain conflicting source intervals and 16 remain unresolved. Nine accepted intervals are earthquake trigger-origin proxies, not measured detachment times. Twenty-two occurrences have both an accepted source coordinate and UTC interval (Figure 3). Completeness is strongly source-dependent: accepted-coordinate counts include 0 of 7 High Mountain Asia frame records, 13 of 20 independent rock-search records, and 2 of 2 web-search records. The small cells and evidence-availability differences do not describe occurrence rates.

Among the 29 occurrences in the registered ERA5 pilot, only 9 have accepted source coordinates, 17 have accepted UTC intervals, and 9 have both; 2 retain time conflicts. For these 9 occurrences, the largest catalog-to-audited separation is 0.677 km and the median is zero. That small conditional distance does not validate the other 20 catalog points: their main limitation is target geometry or unbounded uncertainty rather than decimal precision. Five Denali source-scar digitizations were accepted only after independent map review. Conversely, EarthScope timestamps for Elliot Creek and Pedersen were retained as station arrivals rather than source onset. An independently reviewed USGS second supports Pedersen; Elliot remains unresolved, and two Paat primary sources remain separated by two seconds. The audit thus changes which observations qualify, even where accepted numeric coordinates move little.

5.2 Registered ERA5 temperature pilot

After freezing the discovery inventory, I registered a narrower extraction test before reading temperature at an event location. It selects every included, trigger-time-eligible occurrence with finite catalog coordinates, giving 29 occurrences. These are the original discovery coordinates, not the later accepted source-coordinate set. Hourly 2-m temperature comes from the public temporal-layout Icechunk copy of ERA5 snapshot `T9H8SG2PVXWNY0QNJPJG` [6, 13]. Surface geopotential and land fraction come from the NSF NCAR 0.25-degree ERA5 invariants [7]. Three fixed, non-event probes matched the temporal and spatial layouts exactly before event extraction.

The finite-catalog-coordinate requirement retains 29 of the 53 trigger-time-eligible occurrences, and this input availability is strongly source-dependent: coordinates are present for 2 of 20 records from the independent rock-search suite, 11 of 12 from the landslide-tsunami frame, and all 7 from the High Mountain Asia frame. The later audit accepts source geometry for only 9 of these 29. The summaries below therefore characterize the registered catalog-point subset, not all 53 eligible occurrences or a source-verified subset.

For each occurrence, the extraction retains the two nearest distinct grid coordinates on each axis. The primary cell has the largest land fraction, with distance and coordinates resolving ties; results also retain all four cells. The registered primary window contains the 168 complete UTC hours before 00:00 UTC on the published catalog date. The source audit shows that only 17 of the 29 occurrences have accepted UTC intervals, while 2 conflict and 10 remain unresolved. In particular, the Aru-2 local date makes the original window include three post-onset hours. I therefore retain the registered pre-date result and add a conservative sensitivity covering the seven days ending one full UTC day earlier. Each mean receives a midrank among matched 1991–2020 calendar windows, excluding the event year. This warm-state rank contains secular change and weather. A separate linear-trend-residual rank removes a Theil–Sen line fitted to matched windows from 1979 through 2025. Nonlinear secular change, low-frequency variability, and reanalysis inhomogeneity can remain. Neither rank estimates failure probability because the sample has no non-failing slopes.

The median warm-state rank is 0.586 (interquartile range 0.379–0.828), with 19 of 29 values above 0.5 (Figure 4). The corresponding median temperature difference from the calendar-control median is 0.486 K (interquartile range −0.508–1.414 K). The median local Theil–Sen trend is 0.353 K per decade (interquartile range 0.282–0.509 K per decade). After detrending, the median linear-trend-residual rank is 0.483 (interquartile range 0.267–0.828), and 14 of 29 values exceed 0.5. Selecting the registered representative from each of 25 dependence components gives median warm-state and residual ranks of 0.586 and 0.533, respectively.

In the conservative-antecedence window, the occurrence-level median warm-state and linear-trend-residual ranks are 0.586 (interquartile range 0.483–0.800) and 0.533 (0.300–0.759); 21 and 16 values exceed 0.5. The component-representative medians are 0.621 and 0.552. Buffering changes event-year mean temperature by a median of −0.050 K, but the range is −1.101 to 0.965 K. The

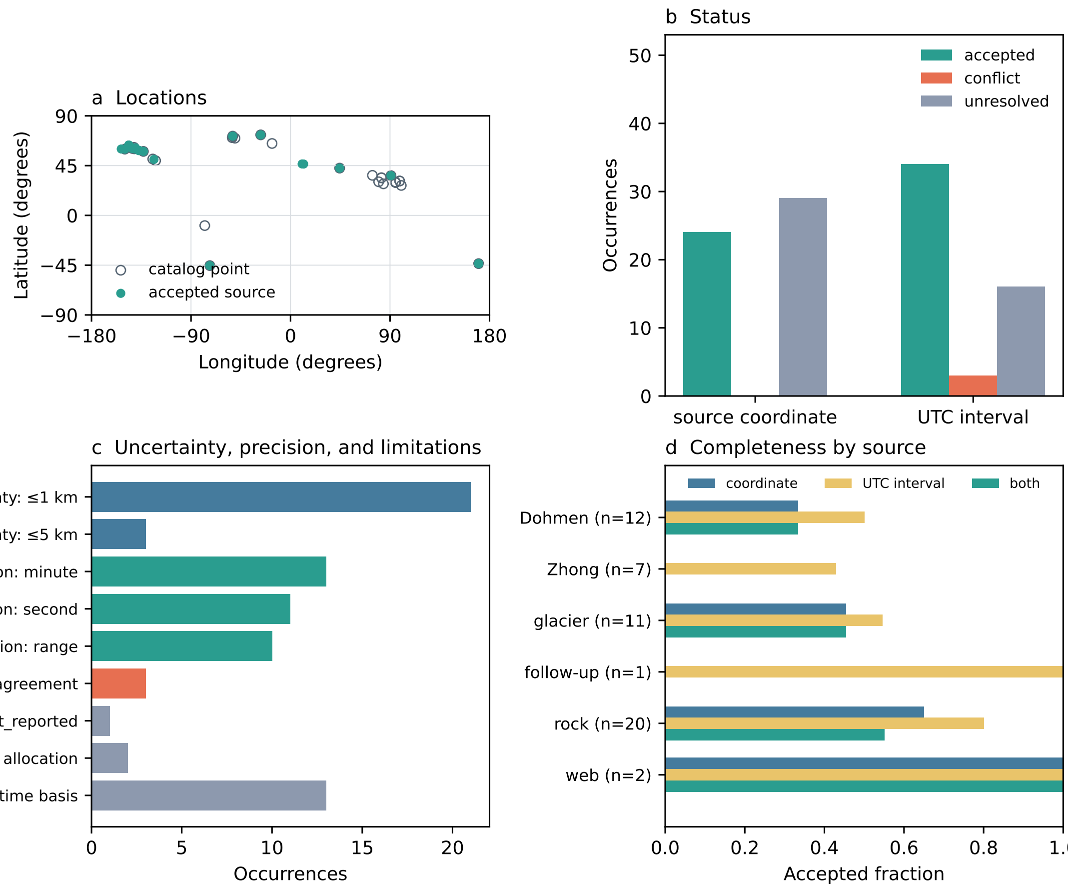

Figure 3: Climate-blind source-coordinate and UTC-onset audit. (a) The 29 finite original catalog points and 24 accepted source coordinates. (b) Accepted, conflicting, and unresolved outcomes. (c) Accepted coordinate uncertainty, accepted UTC precision, and reasons for non-accepted times. (d) Acceptance fractions by discovery stream. Nine accepted UTC intervals are earthquake trigger-origin proxies, not detachment clocks. The figure contains no climate or reanalysis values.

other registered warm-state/linear-residual median pairs are 0.552/0.448 for two days, 0.724/0.655 for 30 days, and 0.533/0.448 for the catalog-date day. The central rank is therefore window-dependent. None of these summaries is a hypothesis test or a separation of weather from every slow climate component.

Grid representation is a material measurement uncertainty (Figure 5). The four-cell model-surface height range has a median of 323 m and reaches 1,489 m at Chamoli. Across these cells, the median warm-state-rank range is 0.133 (interquartile range 0.069–0.167) and the maximum is 0.333; the median range of the temperature difference itself is 0.716 K. The buffered four-cell rank range has a median of 0.103 and the same maximum of 0.333. A fixed lapse correction cancels from ranks because it is applied to both event and control values. It does not cancel from absolute 2-m air-temperature hours above 273.15 K. At model elevation, the medians are 113 of 168 hours in the pre-date window and 121 in the buffered window. At the +1 km elevation-offset scenario, lapse rates of 4.0, 6.5, and 9.0 K km^{-1} give pre-date medians of 42, 11, and 0 hours and buffered medians of 50, 11, and 0 hours. These air-temperature calculations do not imply thaw of rock, soil, or ice, and the offsets are scenarios rather than reconstructed source elevations.

6 Competing hypotheses and discriminating observations

The analysis will evaluate signs and timing that distinguish mechanisms. Debuttressing predicts a deformation change point after glacier thinning or contact loss, with a geometry-dependent lag. Loading has no general sign. For an added load Δq on a planar slope, the simple components are $\Delta\tau = \Delta q \sin\alpha$ and $\Delta\sigma_n = \Delta q \cos\alpha$; the Coulomb tendency $\Delta\tau - \Delta\sigma_n \tan\phi$ changes sign with slope and friction angle, before crack pressure and three-dimensional geometry are considered. Hydrologic weakening predicts velocity or failure probability increasing with water input when drainage is inefficient; hydrologic strengthening predicts a negative relation or seasonal hysteresis as drainage evolves. Permafrost degradation predicts deformation associated with cumulative rock-temperature history and ice-bearing fractures rather than a single warm day. Progressive damage predicts accelerating deformation or microseismicity without requiring unusual weather. Structural release predicts failure along mapped adverse discontinuities. Seismic forcing predicts temporal coincidence with an earthquake capable of producing materially large shaking at the source.

Barry Arm provides a prospective case for checking several predictions. Change points and lags in slope velocity, glacier-contact loss, precipitation, groundwater, and seasonal temperature can be tested with autocorrelated errors. If a glacier term loses explanatory power after detrending or after internal creep state is included, the cumulative 0.99 correlation should not be treated as evidence for a dominant causal pathway.

7 Uncertainty and next steps

Source uncertainty enters at three levels. First, reported intervals are not interchangeable. Chamoli gives a 95% reconstruction interval; Aru reports a conservative linear sum of DEM and post-event correction errors; the probability semantics of some reported Dickson and Tracy uncertainties remain unspecified and are labeled `reported_unspecified`. The Barry Arm depth spread describes agreement among tested parameterizations and time periods, not full inversion nonuniqueness. Second, source classifications can change. The reviewed Nepal seismic record now calls the source a glacial collapse and debris flow, whereas the dated event page retains broader provisional wording [22, 23]. Both records are preserved. Third, some quantities are conditional scenarios. Barry Arm

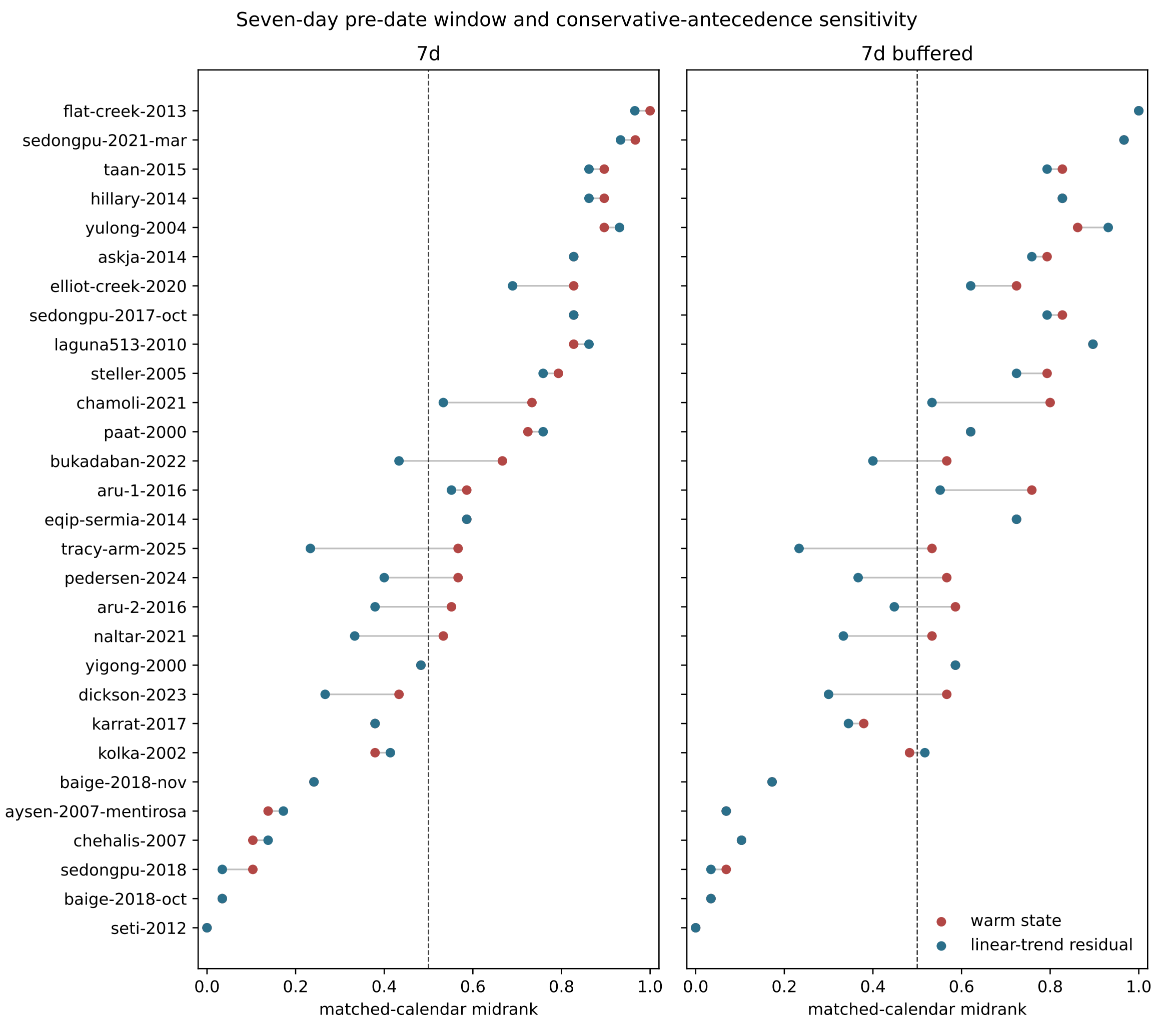

Figure 4: Registered seven-day ERA5 temperature ranks for the pre-date window (left) and conservative-antecedence sensitivity (right), using a common occurrence order. Red points rank event-year means against matching 1991–2020 calendar windows; blue points apply the same rank after removing a robust 1979–2025 linear trend. Lines join two diagnostics for one occurrence. The dashed line at 0.5 is a reference median, not a fitted failure threshold.

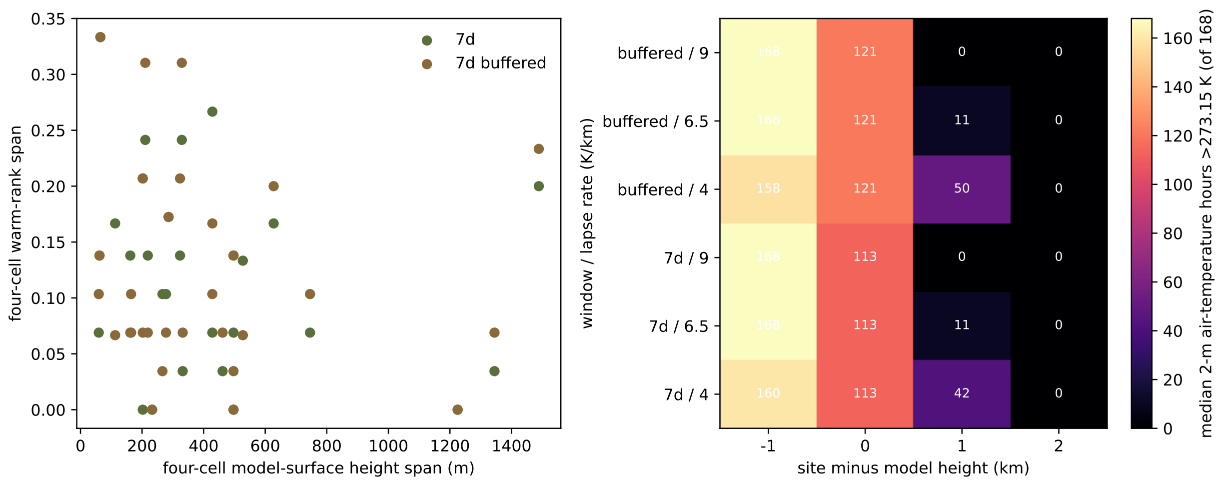

Figure 5: Registered topographic sensitivity. The left panel compares the four-cell model-surface-height range with the corresponding warm-state-rank range for both seven-day windows. The right panel gives the median number of 2-m air-temperature hours above 273.15 K for every window, lapse-rate, and site-minus-model-height scenario. It does not diagnose material thaw, and the scenarios are not estimates of source elevation.

volumes and tsunami heights describe possible geometries, not observations or forecasts of a failure time.

This version completes the climate-blind frame reconciliation, double screening, registered ERA5 temperature pilot, independently reviewed source-coordinate and UTC-onset audit, and the first case-blind susceptible- terrain pilot and held-out susceptible-area convergence test. The audit leaves 29 coordinates and 16 times unresolved and preserves three time conflicts; none is silently removed. The pilot inspected only its registered temperature endpoints and is not retroactively recomputed on the accepted audit subset. A later extraction may use the 22 occurrences with both accepted fields only under a new frozen specification, while retaining all unresolved rows in the reporting frame. The terrain pilot supplies candidate denominator objects, but its three windows do not estimate regional populations and its connected counts remain scale sensitive. The deterministic held-out windows likewise do not estimate a global population, and both registered area strata fail their resolution criterion. A new protocol must first define a physical, scale-explicit representation that passes its held-out numerical checks. A later association analysis may then acquire glacier-contact histories, deformation records, permafrost proxies, additional meteorological variables, and a frozen statistical specification before inspecting those new exposures. Only then can the study estimate whether hazardous alpine mass movements occupy systematically different climatic or cryospheric conditions.

8 Conclusions

The catalog establishes a traceable distinction among initial failure, downstream cascade, climate preconditioning, and immediate trigger. The climate-blind screen retains 73 eligible occurrences while keeping 68 unresolved cases visible. It also shows why reanalysis alone is insufficient: glacier detachments require effective-pressure bounds, while rock-slope cases require glacier geometry, support change, structure, and deformation. In the registered 29-occurrence pilot, central ranks change with the window and fitted-linear-trend adjustment. Neighboring cells change ranks and anomalies, while elevation scenarios change above-freezing air-temperature counts. The subsequent audit accepts source coordinates for 24 of 53 eligible records, UTC intervals for 34, and both for 22. Only 9 of the pilot's 29 occurrences have accepted source coordinates, so its exposure summaries describe registered catalog locations rather than a source-verified sample. Coordinate and time acceptance are themselves source-dependent, and their representativeness is unknown. The blind terrain pilot reproducibly enumerates source-like area, but 30 versus 90 m grids change primary connected counts by factors of 1.3–3.0. In the held-out area test, phase coefficients of variation remain below 0.004, but 90 m departures reach 26.7–36.6% for glacier-contact area and 29.1–37.4% for PZI-intersection area in Arctic Canada and Svalbard; both strata fail. A new physical, scale-explicit representation, detrended seasonal controls, and explicit competing predictions remain next tests. The present contribution is a sampling frame, protocol, and exposure-sensitivity result, not evidence that warming systematically triggered these events.

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